2. Diketahui : Matriks Kunci $A(3\times3) = \begin{bmatrix} a+1 & b & 2 \\ 1 & a+2 & b \\ b & 1 & a+3 \end{bmatrix}$

Jawab

a. $A = \begin{bmatrix} 2 & 4 & 2 \\ 1 & 3 & 4 \\ 4 & 1 & 4 \end{bmatrix}$; $\det A = a_{11}.M_{11} - a_{12}.M_{12} + a_{13}.M_{13} = 2.\begin{vmatrix} 3 & 4 \\ 1 & 4 \end{vmatrix} - (4).\begin{vmatrix} 1 & 4 \\ 4 & 4 \end{vmatrix} + 2.\begin{vmatrix} 1 & 3 \\ 4 & 1 \end{vmatrix}$

$= 2.8 - (4).(-12) + 2.(-11) = 16 + 48 - 22 = 42$

Tabel 9. Minor

b.

M_{ij}	Matriks	Hasil
M_{11}	$\begin{vmatrix} 3 & 4 \\ 1 & 4 \end{vmatrix}$	$(3.4) - (4.1) = 8$
M_{12}	$\begin{vmatrix} 1 & 4 \\ 4 & 4 \end{vmatrix}$	$4 - (16) = -12$
M_{13}	$\begin{vmatrix} 1 & 3 \\ 4 & 1 \end{vmatrix}$	$1 - 12 = -11$
M_{21}	$\begin{vmatrix} 4 & 2 \\ 1 & 4 \end{vmatrix}$	$16 - 2 = 14$
M_{22}	$\begin{vmatrix} 2 & 2 \\ 4 & 4 \end{vmatrix}$	$8 - 8 = 0$
M_{23}	$\begin{vmatrix} 2 & 4 \\ 4 & 1 \end{vmatrix}$	$2 - 16 = -14$
M_{31}	$\begin{vmatrix} 4 & 2 \\ 3 & 4 \end{vmatrix}$	$16 - 6 = 10$
M_{32}	$\begin{vmatrix} 2 & 2 \\ 1 & 4 \end{vmatrix}$	$8 - 2 = 6$
M_{33}	$\begin{vmatrix} 2 & 4 \\ 1 & 3 \end{vmatrix}$	$6 - 4 = 2$

c. Kofaktor $A(C) = \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$

$= \begin{bmatrix} 8 & 12 & -11 \\ -14 & 0 & 14 \\ 10 & -6 & 2 \end{bmatrix}$

$Adj(A) = C^{T} = \begin{bmatrix} 8 & -14 & 10 \\ 12 & 0 & -6 \\ -11 & 14 & 2 \end{bmatrix}$

$A^{-1} = \frac{1}{\det A} \cdot Adj(A)$

$A^{-1} = \frac{1}{42} \begin{bmatrix} 8 & -14 & 10 \\ 12 & 6 & -6 \\ -11 & 14 & -2 \end{bmatrix} = \begin{bmatrix} \frac{8}{42} & \frac{-14}{42} & \frac{10}{42} \\ \frac{12}{42} & 0 & \frac{-6}{42} \\ -\frac{11}{42} & \frac{14}{42} & -\frac{2}{42} \end{bmatrix}$

$= \begin{bmatrix} \frac{4}{21} & -\frac{1}{3} & \frac{5}{21} \\ \frac{2}{7} & 0 & -\frac{1}{7} \\ -\frac{11}{42} & \frac{1}{3} & \frac{1}{21} \end{bmatrix}$

d. Matriks P Blok Piksel(3x3) $= \begin{bmatrix} 120 & (50+N) & 200 \\ 80 & a+2 & (10+N) \\ (30+N) & 90 & 170 \end{bmatrix} = \begin{bmatrix} 120 & 64 & 200 \\ 80 & 3 & 24 \\ 44 & 90 & 170 \end{bmatrix}$

Enkripsi $= E = A.P = \begin{bmatrix} 2 & 4 & 2 \\ 1 & 3 & 4 \\ 4 & 1 & 4 \end{bmatrix} \cdot \begin{bmatrix} 120 & 64 & 200 \\ 80 & 3 & 24 \\ 44 & 90 & 170 \end{bmatrix} = \begin{bmatrix} (240+320+88) & (128+12+180) & (400+96+340) \\ (120+240+176) & (64+9+360) & (200+72+680) \\ (480+80+176) & (256+3+360) & (800+24+680) \end{bmatrix} = \begin{bmatrix} 648 & 320 & 836 \\ 536 & 433 & 952 \\ 736 & 619 & 1504 \end{bmatrix}$

$P' = A^{-1}.E = \frac{1}{42} \cdot \begin{bmatrix} 8 & -14 & 10 \\ 12 & 0 & -6 \\ -11 & 14 & -2 \end{bmatrix} \cdot \begin{bmatrix} 648 & 320 & 836 \\ 536 & 433 & 952 \\ 736 & 619 & 1504 \end{bmatrix} = \frac{1}{42} \cdot \begin{bmatrix} (5184+(-7504)+7360) & (7776+0+-4416) & (-7128+7504+1472) \\ (2560+-6062+6190) & (3840-3714) & (-3520+6062+1238) \\ (6688-13328+15040) & (10032-9024) & (-9196+13.328+3008) \end{bmatrix}$

$= \begin{bmatrix} 5040 & 2688 & 8400 \\ 3360 & 126 & 1008 \\ 1848 & 3780 & 7140 \end{bmatrix} \cdot 1/42 = \begin{bmatrix} 120 & 64 & 200 \\ 80 & 3 & 24 \\ 44 & 90 & 170 \end{bmatrix} \rightarrow$ Terbukti

Apa yang terjadi jika $\det(A) = 0$?

= Akibatnya dari matriks A tsb tidak akan memiliki invers (bersifat singular), dimana ini akan mempengaruhi proses deskripsi citra (tidak ada invers) sehingga citra asli matriks tersebut tidak bisa dikembalikan.